# gpeyt1-47 · Driven pile in sand (β)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

## Problem

A 14 in. diameter closed-end steel pipe pile is driven 45 ft into medium-dense sand. Water table at 10 ft; γ = 115 pcf above, γsat = 125 pcf below. Use the effective-stress β-method with β = 0.38 along the full length and toe coefficient Nt = 40 (from the Fellenius ranges for this sand). Ignore any limiting values. SPT N60 ≈ 22; φ′ ≈ 34°.

**Most nearly, the ultimate capacity Ru is:**

A. 407 kips
B. 260 kips
C. 117 kips
D. 352 kips

## Given equations / data

> Rs = β ∫σ′v dA (use average σ′v of each segment × side area); Rt = Nt σ′v,toe At; γw = 62.4 pcf

## Worked solution

1. σ′v at 10 ft = 1,150 psf; at toe = 1,150 + 35(125−62.4) = 3,341 psf
2. Perimeter = π(1.167) = 3.665 ft; At = 1.069 ft²
3. Σσ′v·L = ½(1,150)(10) + ½(1,150+3,341)(35) = 84,342 lb/ft
4. Rs = 0.38×3.665×that = 117,470 lb
5. Rt = 40×3,341×1.069 = 142,863 lb
6. Ru = 260 kips

## Final answer

**260 kips**

## Why the other values are wrong

- **407 kips** — used total stress (no pore-pressure subtraction)
- **117 kips** — left out toe resistance
- **352 kips** — used toe σ′v along the whole shaft

## References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7.2 effective-stress (β) method, Rs and Rt with Nt

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Jdcs8-i_OA4, https://www.youtube.com/watch?v=hfbcwIJfAjw (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.